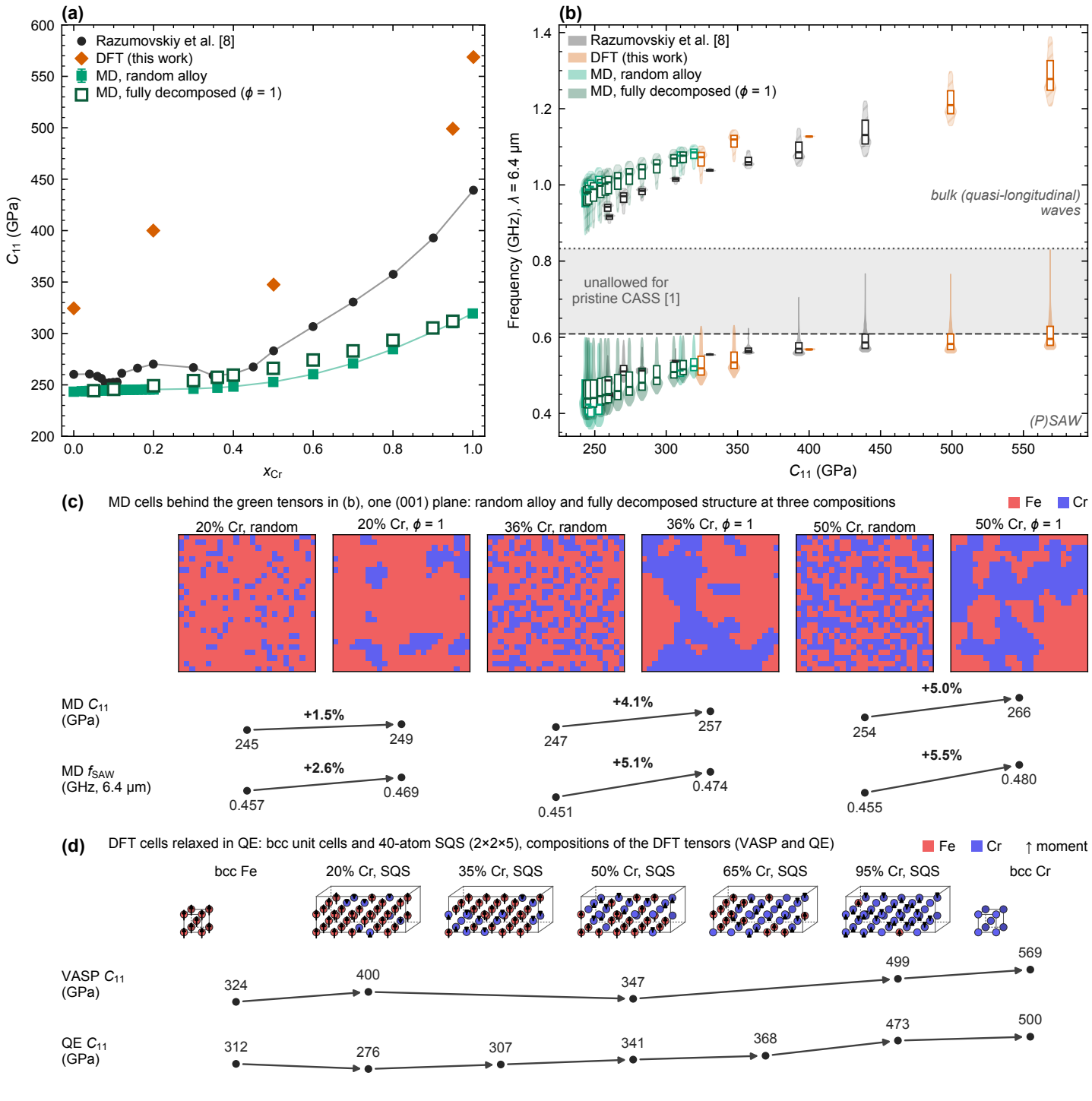

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, fully decomposed (ϕ = 1)
C11 (GPa)
xCr
(b)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, fully decomposed (ϕ = 1)
Frequency (GHz), λ = 6.4 μm
bulk (quasi-longitudinal) waves
unallowed for pristine CASS [1]
(P)SAW
C11 (GPa)
(c)
MD cells behind the green tensors in (b), one (001) plane: random alloy and fully decomposed structure at three compositions
Fe
Cr
20% Cr, random
20% Cr, ϕ = 1
36% Cr, random
36% Cr, ϕ = 1
50% Cr, random
50% Cr, ϕ = 1
MD C11 (GPa)
+1.5%
245
249
+4.1%
247
257
+5.0%
254
266
MD fSAW (GHz, 6.4 μm)
+2.6%
0.457
0.469
+5.1%
0.451
0.474
+5.5%
0.455
0.480
(d)
DFT cells relaxed in QE: bcc unit cells and 40-atom SQS (2×2×5), compositions of the DFT tensors (VASP and QE)
Fe
Cr
↑ moment
bcc Fe
20% Cr, SQS
35% Cr, SQS
50% Cr, SQS
65% Cr, SQS
95% Cr, SQS
bcc Cr
VASP C11 (GPa)
324
400
347
499
569
QE C11 (GPa)
312
276
307
341
368
473
500